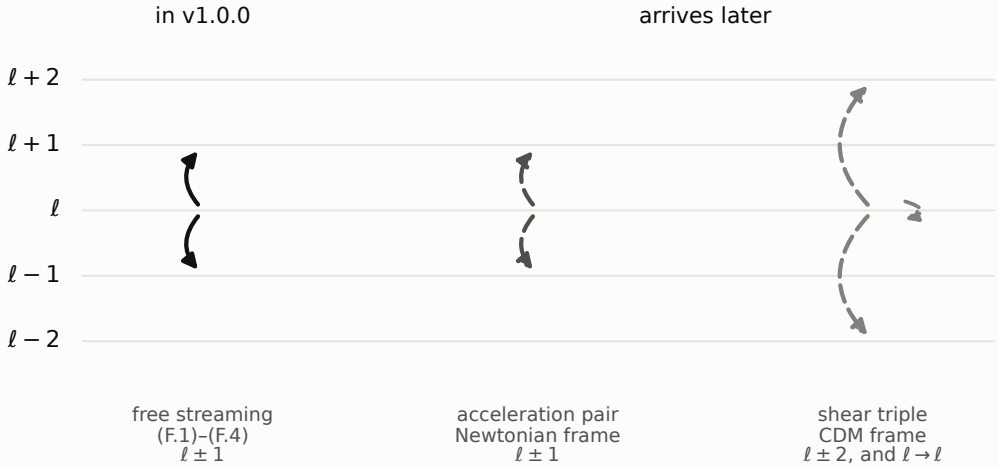
F8 Where each coupling sits in the hierarchy

Solid: what this release integrates. Dashed: what arrives later, and by which frame. The acceleration pair reaches $\ell \pm 1$ because $K^a = \mathrm{D}^a\Phi_A$ carries no direction dependence; the shear triple reaches $\ell \pm 2$ and preserves rank because $K^a = \sigma^a_b e^b$ does. They are one operator in two frames — the frame argument, before anything is computed.



Thomson scattering acts only at $\ell \leq 2$ — the monopole it cannot change, the dipole through v_B, the quadrupole through the anisotropic stress. Every higher rank is reached by free streaming alone, which is why the Appendix F chain is the acceptance spine.